THREE EQUIVALENT CONDITIONS FOR THE DIAGONAL OF A MORPHISM OF ALGEBRAIC STACKS TO BE REPRESENTABLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The statement that the diagonal of any morphism of algebraic stacks is representable by algebraic spaces. Invoked at the start of the counted proof (source lines 419-422). Provides the ambient representability that makes the conditions comparable.
- The criterion for a morphism of algebraic stacks representable by algebraic spaces to be a monomorphism. Used inside the counted proof (source lines 423-435). Links condition (2) with condition (3).
- The description of the pullback of the diagonal along a test morphism as an isomorphism space. Used inside the counted proof (source lines 436-447). Turns the monomorphism condition into a condition on the fibres of the diagonal.

Source reference. Original reference algebraic-lemma-criterion-map-representable-spaces-fibred-in-groupoids: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 431–431. Navigation pointer only: the counted text refers to the Stacks item algebraic-lemma-criterion-map-representable-spaces-fibred-in-groupoids, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original equation equation-diagonal: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 375–375. Navigation pointer only: the counted text refers to the Stacks item equation-diagonal, which is an existing labeled result of this same chapter and is not part of this card. The pointer

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carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-properties-diagonal: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 344–344. Navigation pointer only: the counted text refers to the Stacks item lemma-properties-diagonal, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-monomorphism-loc-finite-type-loc-quasi-finite: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 439–439. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-monomorphism-loc-finite-type-loc-quasi-finite, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-proposition-locally-quasi-finite-separated-over-scheme: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 445–445. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-proposition-locally-quasi-finite-separated-over-scheme, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-section-properties-morphisms: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 248–248. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-section-properties-morphisms, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks which is representable by algebraic spaces. Then*

(1) *Δ_f is representable (by schemes),*
(2) *Δ_f is locally of finite type,*
(3) *Δ_f is a monomorphism,*
(4) *Δ_f is separated, and*
(5) *Δ_f is locally quasi-finite.*

Proof. We have already seen in Lemma lemma-properties-diagonal that Δ_f is representable by algebraic spaces. Hence the statements (2) – (5) make sense, see Properties of Stacks, Section stacks-properties-section-properties-morphisms. Also Lemma lemma-properties-diagonal guarantees (2) holds. Let $T \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$ be a morphism and contemplate Diagram (equation-diagonal). By Algebraic Stacks, Lemma algebraic-lemma-criterion-map-representable-spaces-fibred-in-groupoids the right vertical arrow is injective as a map of sheaves, i.e., a monomorphism of algebraic spaces. Hence also the morphism $T \times_{\mathcal{X} \times_{\mathcal{Y}} \mathcal{X}} \mathcal{X} \to T$ is a monomorphism. Thus (3) holds. We already know that $T \times_{\mathcal{X} \times_{\mathcal{Y}} \mathcal{X}} \mathcal{X} \to T$ is locally of finite type. Thus Morphisms of Spaces, Lemma spaces-morphisms-lemma-monomorphism-loc-finite-type-loc-quasi-finite allows us to conclude that $T \times_{\mathcal{X} \times_{\mathcal{Y}} \mathcal{X}} \mathcal{X} \to T$ is locally quasi-finite

and separated. This proves (4) and (5). Finally, Morphisms of Spaces, Proposition spaces-morphisms-proposition-locally-quasi-finite-separated-over-scheme implies that $T \times_{\mathcal{X} \times_{\mathcal{Y}} \mathcal{X}} \mathcal{X}$ is a scheme which proves (1). $\square$